

# Google Meet for NYC Meetup - 2026/06/24 14:09 EDT - Edited Transcript

## Transcript

**Editor's Note:** This transcript has been edited to remove unimportant content (like our 20 minute break midway) and clarify some wording. Hence, it doesn't exactly match the video recording. The time stamps shown are from the start of the video meeting. The actual event started around the 25 minute mark.

**Dean Wampler:** Okay, everyone. Why don't we get started? if you cannot hear me online, then you probably didn't hear me say this, but you can post in the comments or something if you're having AV troubles. I think most of you may have checked out the QR code on the slide I am showing. There is a web page I put together that will have all of the slides, links to people's projects, a little bit more about some of the speakers, etc. Not everything is up there yet, so check back tomorrow if you don't see a talk that you wanted to download. I decided to do it this way. So, there's one QR code that rules the world, or something, if I can misquote Lord of the Rings.

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Let me start with a little "administrivia". You should have received an email when you registered from IBM that has the Wi-Fi information here in the room. It is also on that little placard on the back desk. The restrooms and the refreshments are behind you in this hallway. Please silence your phones, your laptops, your chest colds, all that kind of stuff, if you wouldn't mind. When you leave today, note that you came in by going to the second floor and then taking the elevator. To lead, take the elevator to the first floor instead and exit that way. And anyone who gets lost has to spend the night in the building. That's just the rules.

I want to walk through the agenda a little bit. We have a bunch of short talks today and I want to frame them in the overall theme that we're covering. First, I'm Dean Wampler, your host. I work full-time in the AI Alliance or almost full-time. Most of the sessions today will be about the challenges of sovereign AI and that word actually has a lot of meanings. I even received a snide remark on LinkedIn about "Sovereign" the other day. "We're taking over the world, are we?" Something like that. But what it really means is that you have agency. You have ownership. You can use AI the way that you need to, that works for you.

Whether you're a government that deals with your data in your special way and you want AI to be tuned for your cultural values, whether you're an industry that needs to tune models to be really good at helping technicians repair cars or whatever. There are a lot of themes rolled up into this idea of sovereignty that we're trying to really support with Project Tapestry. It turns out Yann Lecun talked about Tapestry at a UN meeting yesterday. There's a video floating around of that speech. So, today, we'll start with an overview

of Tapestry and a problem statement of sovereignty from Anthony Annunziata who runs the AI Alliance and is a fellow employee at IBM. He's my overlord. All right, I'm just being honest...

I see three main areas of innovation that are going to come out of Tapestry, in order to support this mission of sovereignty. One is a new form of distributed training called consortium training that's similar to things that have been done before but it has some unique challenges. I'll spend a little time talking about what that's all about. The next area is trying to make AI really meet the needs of everybody in the world. To do this, it has to be tunable and measurable in terms of "cultural values" alignment. So if you live in a predominantly Muslim country, you're going to have a different set of cultural values than we have here and so forth. So how do we support that idea?

For this topic, we will have a talk about a large collaboration including researchers at IBM research, where they are working on the question of what it really means for AI to be applicable to the next billion people or so. This project is not officially a part of Tapestry, but we hope to use this as part of Tapestry's suite of evaluation and alignment tools. Then we have three talks about data. So, you can't do anything in AI without data. You can't train or tune models, implement agents, etc. But the way the world has mostly worked with data so far has been haphazard, just sweeping up data and using it whether you really have the right to use it or not. And there's a lot of data out there that you really can't access even if you want to.

So another thing we really want to do in Tapestry is to fully embrace the data spectrum from completely open to completely private, with the right governance policies and tooling in place, so that we can get the maximum value out of data responsibly and not violate all the concerns that people have about irresponsible use. So we'll start this section with a talk from my colleague at the Alliance, Kaushik Bhatta, who's going to talk about the spectrum of data from private to public and Tapestry's unique approach to working with it to extract maximum value responsibly. After Kaushik's talk, we will take a break. We have some snacks that will be available around four o'clock.

After the break, we'll have a talk from Common Crawl. They will discuss how they are trying to expand their coverage of different linguistic and cultural data sets and environments, to move beyond the predominantly English corpus they currently have. Finally, the last data talk will be from OpenMined, who are experts in technologies for "maximal" control of data use, supporting very careful extraction of information from data while preserving privacy. Ronnie Falcon's talk will discuss ideas for data infrastructure for frontier sovereign AI. Finally, our last talk of the day will be by Alexandra Machado from Red Hat, who will talk about some practical issues of how do you really own and build with AI in your environment, based on what they are learning at Red Hat in projects with large enterprise customers. So with that, I'll hand it over to Anthony Annunziata.

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**Anthony Annunziata:** I'd like to take you through just an introduction to the AI Alliance, where it came from, what we do, where it's headed, and then I want to highlight and spend some time on a really important, very new project, but it's fast moving. It's called Project Tapestry as Dean mentioned. So, the AI Alliance is a nonprofit organization that is dedicated to developing and advocating for open technologies especially at the core AI technologies: data, models, evaluation, and all the challenges in and around the AI stack. It was created about two and a half years ago by IBM, Meta and about 50 other organizations.

It has grown since then and the agenda has continued to evolve, responding to the market. As you might imagine, AI moves quickly. So, our agenda has to move quickly as well. At the heart, what the Alliance is

seeking to do is captured well, I think, in the phrase “scale intelligence outward.” As you may be aware, a lot of the trend in the last couple of years has been in the other direction, right? centralizing intelligence, fewer platforms, like huge scale up efforts and, in particular, many people don't think that that's healthy, not just for individuals but for businesses to try to consolidate intelligence, whatever that full definition of that word means, in fewer and fewer points.

So what we're trying to do is create a better foundation to make sure that every person can have the level capability and control customization of their AI that they want means a lot of different things to a lot of different people and that's the point sovereignty is a really important cross cutting goal of this right the ability to ensure that each person and organization has agency and can make decisions about what and how their AI is used and deployed. The AI Alliance has a community that now includes 205 organizations in 29 countries. We have a lot of activities, including events big and small, including meetups like this, and we have over 100 projects now across the spectrum of topics that I mentioned.

The broader community is about 25,000 and connected to the Alliance in one way or another and this community has been growing in the last year or so, causing us to really rethink the program to be responsive to everything that's happening in AI. Project Tapestry, as you'll hear about soon, is a really important part of that future direction. The program has been built around a few pillars, ones you might expect. There's a lot of activity in and around tools and guidance and ways to better develop agents. There is a lot of work around open data and models. So data sets, trusted data, synthetic data for tuning, domain specific model development. And then there's a lot of work in and around the topics of safety and governance and kind of targeted education around that.

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There's a segmented effort, which is actually in a different nonprofit Alliance organization, a 501(c)(6) that we call the Innovation Association. There we represent a bit more of the industry specific interests and we do some more concerted things including in support of open source. That was on an interesting and pretty stable track for a little while of late. It has become energized recently by some market happenings as you probably are aware. So we're going to kind of roll out some things responsive to that pretty soon. If you'd like to learn more about the Alliance, there's a lot more to check out these links (the QR codes on the current slide). Left is the website and the right is our GitHub.

So now I want to talk about give you an introduction to one of the most, if not the most exciting project that we have right now. It's pretty new, only about, depending on when you start counting, 6,8, or 10 weeks old, but it's moving quickly and it has a very ambitious goal that is going to take some time but we're well on our way and the goal is to create sovereign AI models with “frontier” capability. I'm going to be a little broader than just models here. The backdrop again is something many of you are well aware of, that models today exhibit what we consider frontier performance based on benchmarks and so are basically produced by a relatively small number of organizations.

Those organizations make choices that they themselves control as we've seen especially recently and basically that we have been marshalling around those fewer and fewer choices right so now as you're probably aware we also have open weight models and there are some organizations producing at least near openweight models maybe frontier capability depending on definition but those are also relatively few organizations doing that they're from relatively few parts of the world. Although the weights are often open, the other choices made, such as the pipeline of training and data, etc. are not always open, in

general. So it's really not an open source situation for that either. Now if you think about it though taking a step back we're only really using a very small fraction of the world's resources, especially data, to build these models.

It may look like we're using all the world's data, but no, there's actually a lot more data that could be brought into development if it could be unlocked in the right way with the right assurances and responsible use. There's also a lot of talent that's not involved in the development process. And there's a lot of compute as well and that's growing really fast. In the last year alone, global efforts to build out data centers and sovereign compute clusters have really, kind of caught on fire and that's going to keep going. So what Tapestry is answering is the question, what if we could build a global foundation model bringing a lot of these partners and resources together, many more than any one company can even do, if they're worth \$2 billion or what not?

That's the idea and the way we want to do it and we are doing it is by bringing together a consortium of partners from around the globe that are going to bring together their data, their compute, and their talented people to collaborate on a core platform and a core base model family which will continually evolve. That baseline will be community owned and stewarded by the AI Alliance partners. This means the broader community can build with greater trust and with greater transparency whatever capability they want, by tuning our base models, by embedding our models in some application; they can do as they see fit so that's the big idea. The three goals we're pushing toward first and foremost frontier performance I think one of the lessons behind from a number of the sovereign model development efforts or individual company efforts that aren't kind of one of maybe two or three is that if the model is not really excellent it's not going to get used.

So no one is going to use a model even if it's open source or sovereign or whatever if it's not good if it's not able to achieve frontier capability in an application domain whatever it is right second of course is what we build needs to be sovereign enabling and it's kind of a loaded term means many things but at least it should mean that any person organization or nation can own and control A they understand where it comes from and be the sole determiner of how they use it and the terms under which they use it. The third is actually really important and it wraps together this kind of principle that there's a lot of resources in the world that are not yet being used truly in the model development process that we all use which is structural advantage.

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We're designing this to have a structural advantage that is not easily replicable. Which means if we have this consortium and we have much more data and distributed talent and nodes of compute then that is an amount of resources and the structure of it is set up in such a way that it's actually very hard for any one company to compete with that. So, we're building this not just to have kind of one-time frontier performance and not just to be sovereign enabling, but to be long-term enduring and to be something that the world can continue to invest in and All right. The way that that would work, this is a basic diagram, we'll go into more detail shortly, is in the green. This is what will be the responsibility of the consortium, right?

and the community around it to build out an open source platform that enables a little bit of a new type of distributed training which we'll get into in more detail soon. not quite core training with centralized data distributed compute. it's different. We'll talk about in just a second. And then from that building set a

baseline model set that people can build on and further further train further tune further use. There will be a number of partners there already are coming in partners for data partners with talented people partners that are going to provide compute and lots of people that will then take and use and develop AI that they own based on this.

So just half a level deeper and then Dean I think is going to take us through some of the mechanics of the data management and the actual distributed training and all that. the idea here is that in the AI Alliance we'll have a core operator use that term. There'll be some amount of central kind of orchestration compute and core training. There will be a kind of core team and there will be a fundraising and governance effort that spans across the consortium.

But each partner would contribute to but retain ownership of their resources. It's not a pulled kind of share everything in terms of an IP ownership model. And that's very deliberate. It means that if you're a partner in India or your partner UAE in the US, wherever it is and you have special data sets, you have compute, it remains yours. And in fact, we have a regime of training where you don't even have to actually move the data at all into any sort of centralized training regime. So that's an approach called consortium training. I'll let Dean take you through the more detail there, but it's really at the heart of what we're doing. We're on two tracks. This is going to take a significant effort globally and it's well underway to build out the consortium and to attain the funds and resources needed to get to the scale we want.

In the meantime, we have a focused and growing set of community contributors that are working on three starting work streams in cultural alignment, in kind of the core elements of distributed training, and in cataloging and preparing data to be used in the broader effort. So, I'd encourage you to check it out. And we also have a longer term road map that we're building out. As you might imagine, this is going to take a couple years at least to play out. And so we're building toward that. I'll just conclude with this slide. we launched this effort about let's say less than six weeks ago in a workshop in Paris. we invited 32 people and 30 people showed up with three weeks notice from around the world. So that's pretty cool. since then we have been on a tear. Yan Makun is the chief scientist for this project and it's kind of his passion project at this point.

more than that. and this is his words, so I don't want to put anything into his mouth, but he describes the origin of this idea as something he's wanted to do for a long time, including in his previous employer, and now he's very excited that we actually together now get to do this. we've been making connections around the world, heads of state and companies and all that. So, really nice momentum. Dean mentioned even yesterday at one of the UN events Yan talked about this and there's lots happening around the world. So I invite you to come check it out more deeply And if you're not sold yet, here's Dean. We're actually on time. That won't last.

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**Dean Wampler:** We mentioned this idea of consortium training as a new term for something that isn't completely brand new, because it is a form of distributed training, but it has some unique aspects and that's what this talk is about. So, you might wonder, how is it that your phone can recognize your face? And you would probably guess that there's some AI model there that's very good at facial recognition, but you may not know how it's actually trained. Whether you're using Android or Apple, it's actually done the same way. In fact, what I'm about to show you, I learned from our friends at OpenMind years ago when they got started.

Their founder, Andrew Trask, did some online courses on so-called *federated learning with differential privacy*. I'm basically going to describe it in layman's terms, and I'm doing this because the concepts in consortium training are very similar. This might be a simpler thing to comprehend, to get your head around it. What you may not know is this, you might assume that Apple and Google, and I'll just talk about Apple because they're really kind of the same, just for simplicity. You might assume that they've been uploading people's faces from your phones for a long time and they have some massive data center where they're storing the world's faces and they're training models. That would have privacy preservation concerns and so forth. But in fact, that's not how they do it. Instead, what they do is, they start training a model in some data center and then they download that model to all of your phones. And this is a loop that is going on all the time.

In fact, most of you who are not looking at your phones right now, this could be happening right now on your phone where it does some training on your local data and in general gets better at recognizing faces and then it uploads those changes to the model back up to the mother ship and then the mother ship has an algorithm for merging those changes together. And over time it gets really really good at recognizing faces. and in particular it can recognize your face to enough accuracy that it's a good safety or a security tool.

I have a little homework for you and I should have tried this myself beforehand, but I want you to go home and see if you can get your pet's face to be used as a key to unlock your phone. I doubt it will work because your pet's face looks too different from a human face. I wish I'd thought of this idea before; I need to try this at home with one of my cats. See if I can get it to work.

The other point I'll make about this is we're talking about millions of phones and of course millions of faces that are all working together. And it's also true that as powerful as these devices are, they're not nearly as powerful as a cluster of NVIDIA GPUs. And yet working together they can do stuff like build models. So the advantages are that it does preserve privacy. Nobody has your face data, in this context anyway. The NSA does, but nobody else has your face data. It's all on your phones. Anyway, the approach does work.

The models do get better over time at recognizing faces in all kinds of contexts, different ethnicities, genders, lighting, hats, no hats, glasses, no glasses, etc. And it's also very efficient because Google and Apple don't need to allocate a lot of storage and compute for this process. They're basically leveraging excess capacity that's spread around the world.

This process is called "federated learning", because we're federating these devices, with "differential privacy", the technology for very carefully controlling what information can be extracted from data while preserving other private data and so forth. Really fascinating stuff actually, if you ever get a chance to learn about it.

All right, let's go to the analog now, *consortium training*. Here we have N "sovereign nodes", as we call them, on the order of tens or dozens of them, as opposed to millions of phones. There is a central node, like we had with Apple's face recognition model training. Note that it could be that a lot of the training for foundation models will be done on that central node, or it could mostly be distributed on these other nodes. We'll work this out as the project evolves. Either way, it's a simple concept where I have a relatively fewer number of large nodes. These are data centers in India, France,, places like that and they all have local data as well as global shared data. They'll probably do a lot more training loops locally and then share upstream to merge model updates together.

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We have to solve the problem of heterogeneity that doesn't exist when everyone is using the same Apple silicon. So for example, some of the organizations who are joining are standardized on AMD GPUs, while other groups are using Nvidia GPUs, of course. So there's a lot of technical problems to make this work. The upstream merge algorithm is also more complicated in part because of the longer iteration cycles locally but also because of the diversity of data. One of our forthcoming talks is about local versus shared data sets and you can see that in the diagram here. A really good example of sovereign data would be some data set about the citizens of India that by law cannot leave the boundaries of India. It has to be stored in India. It can't be leaked out to a US data center for example, even though maybe there's nothing particularly private about this particular data set.

So the beauty part of this, which gets to Anthony's point about reaching data sets that have not been reachable before, is that the data center in India could train on that data. Similarly, a data center in Africa could use data on African citizens, etc. Then the model globally reflects these changes that are merged together and the model gets better at these various cultural aspects like ethnicities and distributions of populations and so forth. But there's an interesting technical problem here and that is how do we make sure that the models are not memorizing this data which would effectively leak it outside allowed boundaries. Therefore, evaluation is going to be really crucial. We're going to have a talk in a minute about cultural alignment and that's a part of this broad concept of how do you evaluate that your requirements are being satisfied. This is one of the interesting examples.

Can we guarantee that we're not leaking data inadvertently by doing a technique like this? And this data could be everything from again completely public data sets that have no restrictions to data sets with various limitations like I described.

We call this N + 1 consortium training (or learning; the words training and learning are effectively interchangeable). We are applying core ideas that have been done before in a new way for this kind of distributed training, but adapted to new problems and new challenges.

So the advantages also include another point Anthony made, none of us can have all the resources to build the kind of data centers that are used by organizations like Open AI, but working together we can overcome that resource limitation. The other big thing here, and we'll hear some talks about this as we go forward, we really finally have to get our hands around the spectrum from open to private data and everything in between and how we manage and govern that data so that we get the maximum value out of AI while using it responsibly.

There's an interesting tipping point you reach, and somebody asked me this earlier today when I gave this presentation internally to IBM, if I've got Hindu culture stuff coming in and I've got Muslim culture stuff coming in are they going to start to collide?

When does the model start to break down in terms of really being culturally aligned? And that's the reason among others why you eventually stop training the global model and go to local tuning to really get the performance you want for your local use cases, cultural values, etc. And the great thing is that we want to enable all of our sovereign nodes to be able to take that next step by themselves as they see fit using the stack that we've all developed together. So they don't have to reinvent the wheel or figure out from the ground up how to do fine-tuning, reinforcement learning, etc. Another important advantage is this, once we are building foundation models of this caliber ourselves, because we are building from the ground up

to be as open as possible, with proper data governance. You can really trust the models that you're starting to tune..

Yes, there are a lot of wonderful openweight models, but you're taking a little bit of business risk, not knowing exactly what data went into them, whether it was properly governed and so forth. So, we want to solve that problem. Once you're tuning locally, you can use any data you want that's private for your business or for your company. We can also find the best sweet spots between globally good models and locally tuned models.

This is literally the first issue in the git repo for this problem. I marked it as an epic because it's actually quite massive. But I really invite all of you to dive in any way that you'd like to contribute. There's a lot of interesting problems to solve here. It's issue number 70. And once again, here is the same QR code that was up before.

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Check out the repo, there's a link to it already on this page. All right, I think I have a few minutes. Any questions? Yes...

I think we have to solve that problem. If you didn't hear that online, the question was about is there a supply chain risk when you have all these people contributing? And I think some of the things that have to be solved include how you need to establish trust but verify that all of these sovereign nodes are doing what they say they're doing. There has to be some mechanism for that. There has to be some awareness of all the supply chain aspects, like everything that got "hoovered" into this process. The software used to build the data sets, etc. will have to be eventually covered in some way to meet that need.

There are some people who have joined the Alliance recently, not related to tapestry, but who are very interested in applying *bill of material* (BoM) concepts to AI software. If any of you are interested in that, talk to me and I can introduce you to those folks.

Next Q. If I can paraphrase the question and correct me if I get this wrong but there are other ways in which you have heterogeneity coming into play, such as different levels of privacy, different rules and regulations in different organizations and how they interoperate.

Next Q: For those of you online, not all data that might be available in India has to stay in India. In fact, we've started encouraging people to submit data sets that can be used, which may be in the public domain, but they think that these data sets may not already be part of crawls. It turns out a lot of government data is like this. It tends to be in a very inconvenient formats for using in AI training, so we do expect there'll be some greater data sets collected that don't have the restrictions of sovereignty and don't need to stay local.

We'll decide how we break up the training burden between that central node and these sovereign nodes. It could be that we'll train by sharding the open data or we'll share the same data set to every node and then maybe they train on parts of the model depending on the model architectures chosen (e.g., mixture of experts). Kaushik will talk about it a little bit in a few minutes.

Let's go to our next talk. Now we'll get into this subject of cultural alignment and how to handle that properly.



**Ambrish Rawat:** Hi everyone. I am also from IBM Research but today I'll be talking about this initiative of next billion AI index which is a bit more broader initiative that we've been working on with some colleagues from IBM research and few other folks outside IBM.

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The goal here being that in general there has been a lot of emphasis and lot of incentives that have been given to the industry to develop things in an open fashion and transparent fashion but there has not been enough focus to deliver something for the next billion. There's no incentive structure around it. So I'll get into the weeds of it but just to begin right so one as part of this initiative we conducted few surveys and one of the things that stuck with me is I thought I would state that quote war just so that I can motivate what the idea of next billion index is. So one of the participants mentioned that with respect to AI and AI models in particular they said within our community there is no consensus on what AI actually works for us.

So we don't know how to make an informed decision. and if you think about this carefully, they're not asking about which models perform the best on any benchmark. They're not looking for model supremacy angles. They're just asking given my context my data constraints, and given my sort of users, which technology actually serves me? That's the question they're typically asking. So they want a response to that sort of contextual thinking around model development. this is not to say that performance doesn't matter. I think as Anthony pointed out right the first thing that somebody's going to ask is does the model work or not? Performance definitely matters. That's the first and foremost. But that's not the end of the story and especially for the next billion markets in those settings. So you can contrast.

So sometimes if you look at what benchmarks exist in the model world, they are often addressing concerns around model performance capability performance. So they'll address questions how does this model perform on tasks on code coding problems? Is it well aligned with respect to or have people done safety or security red teaming around it? Is it good at addressing languages? These are all sort of capability constraints. But when a user is looking at making decisions on what model they need to use to solve their problem, what they're actually asking especially in expedient markets is how can I deploy this reliably and sustainably within my application? Where does this model reside? Does it work offline? U will my users trust it? and then with respect to sovereignty, can my community actually govern it?

So there's that difference or contrast in terms of how a model developer thinks when they're developing a frontier model a open weight model but then there's a gap of what the user actually wants. of course it's not to say that there aren't adjacent effort efforts in this, right? There are lots of efforts which address this similar aspects, right? So there are things of course on capability you have all the holistic evaluation systems like Helm which answer all sorts of capability questions and keep track or keep score of the model development. On that front on the transparency and openness side there have been few initiatives.

So one I think worth worth mentioning is the foundation model transparency index which asks the question of how transparent is the development process. It looks at the documentation the governance process behind collecting data behind training and the other disclosures around to sort of rate a particular model and how transparent it is. there is also open weight as I think both Anthony and Dean were mentioning open weight doesn't really complete this end the story of Openness also needs a feedback loop of community actually contributing something back to actually have a say in the development. So the artificial analysis openness index tries to address this concerns on what actually

openness means and iterates both open weight and close weight models. AI progress in general is being tracked by index.

You have other similar efforts on tracking responsible AI commitments whether they're implemented or not how good a government is but none of these actually track utility and adoption for next billion and that's the gap we're trying to address so we've been thinking through and pushing this initiative of next billion AI index in which we have categorized three sort of broad themes around which you mentioned different dimensions which are relevant for the next billion folks. Now the first one being effective efficiency. So this addresses the question of can the system actually deliver cost value per dollar right? So I think you must have seen debates around token maxing being a problem these days and people are asking for what's the maxing similar question right so what does it actually give you value in terms of performance and resource requirement per dollar that you put in.

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It also incorporates energy dimensions and then infrastructure dimensions. Second and perhaps one of the very important ones is operational practicality. This talks about whether or dimensions within this talk about whether the AI system be Can it be adapted for the contextual local needs and I'll get into the subdimensions later on. Third is societal integrity. That's the part where I think it's often not focused or not as emphasized but I think for a sovereign development that becomes extremely crucial is whether the system that you're developing is actually faithful to the value system in particular local context. Does it allow for some sort of shared stewardship.

So within these three buckets we have out of 10 dimensions it's very hard to capture all different holistic dimensions within a small framework but at least trying to collect them in sort of one framework helps you think through the relevance of these dimensions. So with effective efficiency we sort of think of two important ones. One is as I mentioned cost per dollar. So that's on the performance angle. Second is resource efficiency. So that's like how easy it is to host certain things. how challenging is it within a particular local context to size inference second is operational practicality. So within those we have five different dimensions. so I already mentioned the customizability Second which is I think often very important is interoperability. Can it actually work with different tools?

Third I think a very important one u which covers a lot of different aspects of safety security and just this general robustness to noise is this idea of resilience reliability and robustness that covers it's sort of an all and passing one with respect to robustness usability and automation education are two other ones u with usability what we mean is how easy it is to get up to speed and start running these things is there enough documentation aail available that is in the education front. does it in the few things which are say API based often shine well on education because there's lots of online material collaboration communities around it that makes it easy.

And then finally around trust sorry on societal integrity there sort of three broad themes trustworthiness and ethics these are overloaded terms they mean a lot trustworthiness within whether the user actually trusts the developer on their development process they need to be open about it they need to make sure that they're addressing the local values so there are three important dimensions here trustworthiness inclusivity and pluralism and then finally openness and collaboration okay

But what does it actually mean to use such an index in practice right now I'll mention why this index differs from the other ones in the next slide but just to get an idea of how do you actually operationalize

this so broadly three steps right so once you want to figure out whether your local system or let me set up a scenario right so suppose you are a company who is trying to figure out the best model that is useful for logging in a system. So if you have a classic logger in an operation exercise in a business automation process or in an industry, you have some field workers who want to lock certain things for day-to-day. You may want to use certain speech model, right?

classic speech model translate voice to fill the form or fill some fields within your tool. So ideally for that use case you would want to figure out what's the best model that delivers value to me. there are multiple options available. There are a few open APIs available. There are some small models available. Typically you will begin the exercise by first listing down the solutions you want to compare and then for each of the dimensions you had mentioned you would go through the exercise of defining what's the best for that dimension and what's the worst define a ladder this gives you sort of a rubric we did this exercise for model classes but I think you can be more specific than you have to make an active decision within your business process once you've decided

on sort of a ladder. I think the next step that you undertake is build on some existing benchmarks which might exist for certain dimensions. just to highlight I think resource efficiency people have measured there are multiple that exist. There's the ones that are gaining some popularity are things like AI energy score or intelligence per watt from Stanford which has one mechanism to quantify what resource efficiency could mean. So you collect the sort of existing benchmarks and then for the ones which you don't have benchmarks you actually engage with a local practitioner to make sure you have a common understanding of what weak and what strong means on that front.

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I think the third step is actually a very important step which is something which we overlook and the third is you need to design this thing with the local stakeholder and that's where it gets tricky in some phase in some ways but also grounds it more in a local sovereign context. So when you design with local stakeholders you don't prescribe what the right and wrong is for that dimension. you ask and you work with them. That's the difference from other indices which we're trying to highlight with the next billion index. Why do this exercise? The reason you want to do this exercise is to surface trade-offs. And this is just one example of how this is indicative but this is one example of how trade-offs may look and just this is probably intuitive for some of us.

For example, a lot of closed models they may perform very well on benchmarks but they may not be very efficient when it comes to cost per dollar depends on the pricing model. in terms of operational practicality that is like are they easily usable? They may not be right because a lot of them are fixed. We don't have options to fine-tune adapt for local data sets. on the integrity of course they're closed in all forms. They're pretty poor. on the open weight models where performance is an issue. They may not perform the best on benchmarks but then their cost per dollar especially if all the hosting is done locally then at least that part is well well accounted for.

They are more adaptable although they also have challenges in customization and adoption because you do need some resources to adapt right then it's not like you can adapt them without fine-tuning without training some local adapters. but they perform really well on u or intuitively they'll outshine at least on the societal integrity part because a lot of control is in the developer hand. a good sort of sweet balance often is when you have somebody who can combine these open weight models with some sort of basic

orchestration framework where the logic is not just in the model but logic sits outside and you are able to leverage tools like lchain and a few other existing tools. but it's just to give you an idea of what trade-offs will surface when you start to implement or apply some index of this form on your design decisions.

So we did test this whether these dimensions make sense or not. So we actually went to the developers in the next billion markets and did some surveys and asked some product managers within different geographies from mostly predominantly in India and Africa on whether these dimensions make sense to them and just to highlight one quote I think overall the three buckets make a lot of sense to them and cost effectiveness was definitely the one that they highlighted as the most important

with usability as the second and user and trust as the third. So broadly the three clusters do resonate well with them. So we did this exercise with the developers to define these. they also mention few places where so they'll find it useful in comparing technologies. also just as a digest if somebody has done the research then these things do surface trade-offs which is their day-to-day exercise. So if something is readily available on this front then those decisions are easier for them. they did mention a few other challenges in terms of how effective next is. So I think as developers a lot of them do prefer another level of detail but as policy and decision makers they have another angle of what is important what is not. So we're trying to refine this in the next exercise.

But I think overall we had quite positive sentiment. so so far I did shame benchmarks for not performing or not giving enough information on usability but am I just coming up with another index? So I'll address that up front. Right? So this is not just another index for three different reasons. First the idea here is not to come up with a leaderboard. It is to come up with an incentive structure. this is to replicate the exercise of what in some fashion in the same spirit at least as what Stanford's transfer index is which is some way of making a measurable mechanism to capture utility and adoption for next markets which is what transparency index is for transparency we want the same effect and so that the developers have an incentive to deliver to them as opposed to just caring about supremacy second more important thing this is

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not a universal score of usefulness and I would like to emphasize this right so I think as you noted in the third step of how you operationalize this you have to design with the stakeholder so you can come up with the sort of first stage diagnostic to realize which dimensions are important but you do need to involve local stakeholders to actually figure out whether a particular model is delivering value for them. So make sure that all the rubrics are defined for use cases. This is not universal score. The rubrics will use The dimensions and their priority will be use case specific. And we noticing this as we're talking to the sort of next set of case studies. You're noticing this pattern that there is a priority to which dimensions are important for some use case which may be important for some other use case.

And also in third form I think it's to be taken the same spirit as say even sustainability goals right they don't rank a winner we don't want to rank something some nation as a winner or some developer as a winner which is outshining because if you go in that line I think all indices are gameable in that fashion all benchmarks are gameable so we don't want to repeat that exercise and I think Dean pointed out right that often this last mile of whether a model is actually going to be useful in a sovereign and a tapestry structure

is when you actually fine-tune it for your own use case after having learned something in consortion training that's relevant at the base level but you do apply some techniques to fine tune it for your case in the same way make sure you have to define the rubrics for your own case right so what does this mean for sovereign AI right so there are two sort of things I'll highlight for the two markets that we have looked at so far one is especially in the next brilliant context one is the Indian market and the other is the African market in both of these if you look at the government strategies or if you look at the surveys that some of the think tanks have done.

So in India I'd highlight the one which is from the budget from this year was also highlighted. A lot of these points were highlighted in the AI impact summit in February that India wants to shoot for a sort of small model strategy open strategy and not focus on model supremacy as the only path to success and you need to define application specific AI build things which are more useful for local context. So the same sort of resonance with the themes we have similar things in Africa if you look at their continent strategy and there are versions of this for each country. We looked into some in Nigeria and Kenya. they have similar sentiments that they highlight that it needs to first solve the problems of local challenges in Africa which is around sector specific things like healthcare,

food security and not just intelligence for the sake of intelligence. and they also have govern governments within the nations as sort of the first class adopters of these technologies. so they want AI to leverage so the themes that emerge out of this right so I think in sovereign AI just because you have ownership in terms of hosting and contributing to the training and data the story doesn't end there. Utility is the next step. So something needs to be like a nation can build something which is all sovereign in the sense of how data was shared but if it doesn't address a user's problem it doesn't do the last mile then it's not useful. so it needs to be deployable that's the other thing. so a lot of benchmarks if you replicate will not be measuring this. So we want next sort of things to bridge that gap.

So broadly the way it fits within the sovereign AI narrative is essentially in that last mile of measuring utility and adoption for the local markets. just to finish off I'll highlight sort of a broader paradigm that is more specific to IBM and the way we've been thinking about the use of AI and LLM in general is the notion of so with generative computing what we mean is a new shift in the way computing is modeled where instead of treating LLMs as these entities with a unreliable behavior you try and build a robust system around it which treats as just programmable components typically how you do in software. So you're not prompting but you're programming around LLM.

what this results in is a more composable stack where you can combine different modules. You can have adapters, you can have certain agentic operations etc. But you can combine them with explicit boundaries help in maintaining sort of secure taking care of security concerns so that data is not inadvertently taken as instruction at any stage. but if you do this a bit more formally from top down right where you start orchestrating things around an operating system structure then what you get is a much more reliable software system than just unreliable chat bots. and that's sort of the play that we've been pushing towards or as a new paradigm for computing.

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And a lot of the themes within generative computing resonate with next packs and I think just by design right so one thing generative computing does say that you don't need to use a truck for all your last nine deliveries right you can do certain things with small models which is directly related to say some sort of

resource efficiency constraints that I highlighted within the first cluster u modularity and adaptability are similarly related so maybe for certain specialized tasks you can use some sort

PET law adapters to do certain classification tasks. You don't need a whole frontier model scale to do such small problems. so that takes care of the second dimension. and then of course within IBM strategy, we've been releasing all of our models open source. So the open weight strategy is at least one step in the openness and collaborative sort of framework. doesn't end the story. You do need a feedback loop, but at least it's a step in that direction. And to that end we have two sort of projects which we are pushing. One is Millia which is sort of the manifestation of generative computing how it would look like in practice which is an open source Python library that programs on top of LMS. And the idea of granite switch is this idea of modularity built on top of small language models where you are able to reuse certain KV cache for improved inference. of course there's a lot of technical depth.

There could be probably another 20 minute presentations on each of them but broadly they do address some of the concerns that I highlighted within XP packs. so we would invite you to take a deeper look. there's a technical note on it available in archive and of course this is an ongoing excel. It's an initiative similar to AI Alliance and tapestry where we are engaging with different partners to see the value and refine the value of expats itself.

**Dean Wampler:** So now we start our three talks on data.

**Kaushik Bhatta:** First off, thank you for everyone, we appreciate you joining the Alliance here in New York. We have quite a big community across the globe, especially in San Francisco, but we're going to be building up our community here in New York. For a little bit of background, I'm a partner at the AI Alliance. As Anthony mentioned before, we have several hundred members across enterprise sovereign governments. So, if you are not a member, please reach out to me after. That's my only spiel. So, we can get you involved with a lot of our efforts. The topic we're about to talk about is pretty important. And there's quite a few proof points that also interject into what we talk about in terms of data.

When we talked about data generally and we talk about closed models, they're not exactly compatible and that's what we'll start to address. But before we really get deeper into it, one thing I want to talk about is we're already in a state of affairs across the globe where frankly AI is getting booed in commencement speeches and people are pretty upset about what's happening with data centers. And a big part of that is we naturally realize it's not just about being a good person or good leadership. It's about systems and it's about incentives. So I think we need to also take a step back and ask we're in an incentive structure right now where just by earnings reports we're looking on the order of \$130 billion spent this last quarter that we actually have data for that's grounded in data center capex. So the amount of compute talent data and capital at this scale it's just a monumental effort. So what does that mean for society?

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And I don't think we talk enough about that, about open data, what it's actually going to mean across, many different areas of our industries. And we've already started to see some glimpses of this where chat GPT is starting to enable ads with people trying to figure out new ways of essentially capturing different parts of the value chain. And that's the kind of error that we're in. So with closed winning, which is the case right now, this is something we're not yet fully addressing. And I think we need to pretty much have a better talk about what that means for society. one implication is if you take a look at there we go what's already happened in the common corpus. Last year common crawl was here and I think we have some of

them here in the audience and they talked about what was happening in the early web we had robots.txt which essentially gave permissions about what data could and couldn't be scraped.

we've already crossed past that and people have started to take decisions on what data they want to be able to enable through the website. not a lot of people necessarily listen to it, but even when we start talking about public data, there's an exhaustion window that we're going to come up to with our existing models. And I think we also need to look at the whole scope of data that actually exists across the globe, which is a lot bigger. so even with the amount of talk that we've had with closed models and the amount of data that they're starting to capture, it's a fraction the amount of data that actually exists. So we talk about that and the common corpus soon as this thing loads. There we go. So we're already running out of public data in the commons. a lot of it's now being restricted by terms of service. Some people listen to it, some don't.

There's a lot of difficulties in terms of enforcement, but I think we need to also take a look at the amount of data that could be unlocked if we start to think about the open technologies that we can enable in terms of the architecture. And I apologize for that. There's a one minute gap almost in the slides appearing, but it should appear in a second. There we go. So if you think about the amount of data that's actually out there and this is a logarithmic scale, there's a lot more beyond the open web, If we even think about one hospital per year, you're looking at the order of pabytes. if we think about genomic sequencing, it's significantly more. If we think about IoT and machine telemetry, The amount of information we can get from the nodes that are out there, it's a lot more information that's captured between enterprise, other resources, individuals, global healthcare, significantly more.

And we're looking at a lot more information that the world can capture if we have better technologies to create that infrastructure. And that's not something that we can necessarily unlock with closed systems today. So a big part of tapestry and the value proposition is the ability to unlock those types of data sets. And they exist across different spectrums. And I think that's one of the things that many practitioners are thinking about this. They're exploring what to do today. how they can build more security around it, what they can do with existing architectures as context windows have opened, but it doesn't truly unlock a lot of this information because of regulatory concerns, a lot of things that happen inside of commercial enterprise. And I think the way in which tapestry is starting to go about developing out an infrastructure for training unlocks a lot of those.

So this has happened when it comes to the open web and the kernel. this was one of the things that Yan Lun brought up at our Paris workshop. He started with a little premise trying to bring us back to the early days of what Linux brought about. So if you think about the number of devices that are out there just operating systems that are closed and a little bit of an aside because I think this is kind of an irony and the universe has been quite humorous with us recently. did anyone see the little thing that happened in space with Windows and Outlook by chance? Okay, so there was a show on Netflix. We're getting the name off hand, but there was a little snippet where they talked about trying to intercept a rocket coming back into the atmosphere. And ironically, as they were trying to do that, Windows needed to update, which is kind of irony, So, ironically enough, we had the mission out to the moon that went around the moon.

And as they were doing that, not a joke, we had the folks, the astronauts on board have to essentially phone in, I believe, to Houston and ask for their outlook to be looked into because they couldn't get email messages back from command. That's the world we're in. So when we think about closed versus open architecture, there's real unlocks in terms of what it means for the world. All of the major supercomputers, a lot of different things were unlocked by Linux, by Unix. And the web itself having protocols that were

actually open allowed us to have the framework to build websites to build the kind of infrastructure we have. We're not talking about that enough when it comes to AI. So when it comes to incentives, open source is not just asking for better people, better leadership, it's building a better system. We've had this unlocked across different stages across technology.

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It's one of the most ambitious things that humanity can necessarily do is to essentially combine to contribute to a common corpus. But these things has pushed research, and has pushed technology to new bounds. And what could it actually mean? So, one of the big things about the experiments that Tapestry puts together is the fact that if you think about it, only about 5% of the world's population speak English as a first natural language. yet 50% of the web corpus is in English. Yet we're looking at a world that has roughly 7,000 languages. So for people around the globe, this is visceral. They're seeing their languages possibly disappear. They're seeing their culture and norms not be reflected in existing closed models. And you have a lot of unlocks when it comes to enterprise with dark data, information that's step kept inside of commercial enterprise and clinical data that goes unused. And there are some examples of this already appearing.

So this is marketing. It's not necessarily peer reviewed just yet, but I've heard of the examples of Palantir going into hospital systems, unlocking new capabilities with the amount of data that they have inside of some hospitals and reducing sepsis deaths. that type of information exists in many different settings, but we're unable to unlock that without the infrastructure of compute and an open infrastructure to be able to utilize that information. in terms of research, there's a lot of inbound that we've received from different organizations that want to unlock data that they have that's decaying. And if we can build an open architecture, it allows that opportunity. one of the things that was brought up before in terms of an open question is the kind of regime of data, right? So there's regulatory risks, there's cultural information that's not being reflected, and there's commercial interest of things that cannot be shared.

So this is a big part of the things that we have to think about. So that means in some instances data may not even leave a specific data center. in other instances you're talking about cultural data that's not being reflected in a closed lab because they made the decision this is not a frontier mix. in other instances for commercial enterprise they can't share that with other entities because it's sensitive. So these are really hard challenges that we're navigating around and by having an open architecture of members that are contributing to this initiative. We're looking at different ways of essentially unlocking this. So one is the spectrum of data that we're looking at data that can actually stay with the node itself to be able to meet the regulatory conditions. So this is sovereign data things like military sensitive data, possibly hospital data, etc.

But that's a bit more long-term. In our current initiatives, we're looking to train with these nodes similar to what was brought up before the HubSpoke model to build a central model that govern. We're also looking ideally to have fully open information that we can unlock in release if we're able to build that common corpus of infrastructure. Dean kind of touched on this but this is exhibited in some of our P1 milestones. there has been different federated learning architectures and algorithms that have been used. We're testing it out ourselves. So just an honest caveat right now to build at the frontier scale there are significant research challenges that are going to be presented and new scientific unlocks in terms of how we proceed with this.



But there have been initiatives to help us develop better structure to have federated learning for training. but these are things that we're going to be going up against as we start to put this consortium together, which will be fun. And proof points beyond this just being a simple initiative that we bring out to the world and just talk about as in theory, we have people that are starting to buy in significantly. So with the unlock in terms of sovereign demand we already have multiple models around the world that are looking to contribute to our initiative and we just recently received a quote specifically from the principal scientific adviser to the government of India specifically saying that India has long believed that science advocates fastest when it's open and shared.

tapestry which lets nations build frontier AI together while retaining sovereignty over its data and models reflects the collaborative self-realigned innovation India is advancing. So we've gotten a lot of unlocks from different sovereign entities including government that want to support tapestry. and the reason for this is quite evident for folks. It's about their value systems. It's also about being able to have infrastructure that they own. Now, I wanted to kind of rest on this slide for a little bit and kind of give a little bit of something I didn't put in the slides, but I think gives you a little sense of a lot of initiatives of folks that are kind of joining and helping us. And I go back to the last one to talk about this. who here out of just curiosity actually sells outside of the One. That's it. I expected a lot more. I know Anony's up. All right. Good.

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The reason why I bring this up is because there are a lot of things that we're not actually as privy to about what's been happening globally and this has been an incredibly trying time. So, if you talk to folks with our organization, we have several hundred members. we just did a summit during the India AI impact summit. And we have folks that are selling to sovereign nations across Australia. They have enterprises around the globe that they're working with and a lot of things have changed just in the last several months. if you're looking at commodity prices that just went up 40% overnight, if you're looking at how we relate to different allies, people have to make a real hard decision about the tech stack that they're implementing. And what they're thinking is this could be used essentially to turn them off.

And that is a massive risk you cannot take if you're a sovereign nation outside the US. Who here has actually heard of the 2017 cloud act? Anyone? So, a lot of initiatives around the globe that have been implemented by kind of our regulatory regime lead to different sharing of information that they've had to develop different frameworks around terms of how they set up their data centers. That's not something we talk enough about in the US, but right now with the recent kind of advancement of what just happened with Fable, this is a visceral problem. Technology is not just a nice to have. It's not chat bots. It's not just social media. We're talking about government infrastructure, voting, healthcare, right? Finance, money transactions.

That type of infrastructure when it starts to enable into AI is absolutely critical. So one of the big reasons why we've been able to get advancements with different governments around the globe is because they realize they have to have an open infrastructure that they can own. It cannot just be rented. So Tapestry itself is pretty ambitious. we're lucky to have some fantastic people supporting the actual endeavor like Anthony, like Dean, Yan Lun that's out there speaking all the time. I don't know how he has time for it. but we do believe that this is a real groundbreaking moment for us to embark on a mission that's just as ambitious as the ones we've had before. And it's something that we feel, frankly, is absolutely necessary

for this time. The inability for us to have an infrastructure that we can enable around the globe, that is open, that other people actually have agency into is a categorically bad decision.

And we're not talking at all about the implications of what that means in science, in technology, for our society, civil rights, but that's the kind of thing we need to be thinking about. So tapestry itself itself being an open unlock and the amount of data is kind of the market capabilities. Frankly, this is the kind of era where we need to unlock civilizational level funding because it's going to unlock the next civilization. with that who want to join tapestry there are a lot of people that are joining already foreign governments around the globe different sovereign models different organizations that are represented in the audience as well as across our network and please definitely join GitHub by Dean any questions

Please.

01:40:00

Yeah. Absolutely. let me just repeat a little bit of the question to paraphrase for the people that are virtual.

It's about the gap between the regulations themselves versus how it's interpreted by AI. and implementing the fact that even if you touch some of these data sets, it can release things that should be private. it's absolutely a big concern. I don't have a full answer for it, but I have a theory and thesis of how I would align for it. It's kind of ironic because I think some of the best thinkers have been thinking about this for decades, including IBM. So in the 1960s they said something to the effect of management decisions shouldn't be made by a machine because a machine can't be held accountable. I have a theory about what's happening. I think human beings themselves today are still being held accountable. And what I mean by that is that individuals themselves are using the AIs, they're using chat bots, they're using different frameworks because inevitably if that decision is made or that word goes out, it's from you.

And a lot of the regulatory needs are also about enforcement. So it's hard to hold the model accountable. I think the thing that this does unlock is the nodes themselves being contributors into the tapestry project means that they have liability, That's a little bit of the framework. it's a lot deeper than that, but I think we're going to be dealing with those challenges for the time to come. But I do think that's a part of the way that this makes they have the choice of how to essentially contribute. And of course we're going to help them through it. But I think that is a bit more of the alignment between the liability versus who's actually contributing information and how that be received because to your point it doesn't even end there. It gets even harder. The regulatory differences between these different areas. even in the states it's different between states.

is different between cities and in our workshop we had interesting talks about because these were sovereign model developers primarily about the fact that even the same question can be interpreted very differently between different areas right because you're also talking about moral equivalencies that's a philosophical debate yet that can't be reflected by one closed model that's hitting AGI right let's forget about that we need this to be reflected by different regions so I think there's going to be a lot of different interesting ways of going about it, but I think the ability to apply that to the people that are contributing to tapestry is probably the right method. I think Deian and Anthony may have better answers in terms of how they've been thinking about that as well, please. Yeah.

So for the audience on Meet, we were just being clear about the fact that this is going to be very tricky as we start to figure out even our training capabilities and algorithms in terms of what stays private, So the

way in which we're going to do that is frankly by having different experiments as we go and progress and we'll start to have a lot of discoveries in that process. Please

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Hallelujah. No, I think that's a good point. I will say the thing I like about working with a lot of researchers is they question absolutely everything. And I think we're going to test a lot of different things to prove, quandries out.

And I think, to Anony's point earlier, it's a little bit down the line to prove some of these core concepts and prove that, we have privacy preserving architecture and our training architecture, how the nodes contribute for us to get to the point of a lot more sensitive information like healthcare data. So, we'll discover that along the way, but we'd love to, find out more of how folks are dealing with this. I'll also give an aside, again, not in the slides, an aside. enterprise licenses, people signing it. I've talked to a lot of folks, CTO's of major banks, financial institutions, and other groups, and it's one thing to sign a different enterprise license. It's another thing what data do you actually expose it to, Because in the end, it's your liability. And I think there's a big disconnect there. And for us actually building this architecture, it's about what people are going to actually contribute. and we're going to start to discover a lot of that as we start to go along. and it goes to one of the first slides. It's about the incentives being misaligned, right?

If you are seen as a consumer of this with the amount of spend that's been out there, we don't exactly know how that product roadmap is going to change to be able to capitalize on it. And that's one of the biggest fears of those unlocks. Please

Yeah. Yeah, it is.

**Dean Wampler:** We'll take a break till 4:20 and then we'll resume.

02:20:00

**Pedro Ortiz Suarez:** So, yeah, I'm principal research scientist at the Commonwealth Foundation and thank you so much for the invitation. Very happy to be here. Today I'm going to talk about expanding cultural and linguistic coverage of common crawl. So first a very quick very brief introduction of common crawl in case you're not familiar with us. So we are a nonprofit organization.

We crawl the web and make the data accessible for programmers and data scientists and researchers. we are a free archive and have been since 2007. We were funded by Gilbas. our data is hosted on the open data sets Amazon web services and I mean I have to update that one but we have been cited of over 10,000 research paper I think it's 13,000 by now I mean at the beginning common crawl was more targeted towards building stuff like search indexes back in the early 2000s but over the course of the use people have used it for multiple

applications such as well the things you see over there social political research linguistic analysis especially corporate building in that one internet security but of course I think most of you would be familiar with a IML training application but there's a lot more stuff that our dri users use our data for so we're speaking about languages and cultures so one thing that is interesting to look at is the distribution of downloads by geographical location. So as you can see in this picture North America is very well covered Europe as well and well to a certain extent Asia in countries like China and Japan.

But then when you look at the South America and Africa, you see that the downloads are significantly less than what you see in North America and in Europe especially. something that is important to talk about I think is that we are a polite We crawl slowly. we slow down when we encounter errors and we do it with exponential backup. we always respect robots exclusion protocol and also metatags like the one that you see there. Our crawler always identifies itself with CCbot that's the user agent and on our website you can find a list of fixed IP addresses.

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so that you can verify that it is indeed us that is crawling your website. this is because we have seen crawler impersonation to a certain extent. So to the main topic today, let's talk about our multilingual initiatives. so to motivate that a little bit why are we interested in this? this is because this is the current distribution more or less of the languages covered by common crawl or that are present in common crawl. so as you can see I don't know if you are able to read it but now a little bit more than 40% is English. actually in recent months we have managed to push it down a little bit to 39 38%.

But the next language which is always either Chinese, Japanese or German you can see that they drop significantly to 4%. So as you can see at least at the web scale every language that is not English can be considered under represented to a certain extent. So we have two ongoing efforts right now. so the first one is to expand the coverage of our crawl and this is for languages but also cultural and I will explain how the cultural part works and we are trying to do this for languages other than English. So we use this lo acronym and that's the first initiative.

The second initiative is improving language identification because this allow us to actually monitor the coverage of the crawls and make decisions at crawl time to actually crawl a little bit more of those under represented languages. So with initiative one what we are trying to do is to crowdsource seats. we are doing this with the community and we ask them to share some of the URLs of the websites that they visit so that we can better represent their languages, their communities, their cultures.

The idea is that we will inject this URL seeds into our seed crawl and then compute a web graph that uses harmonic centrality to better rank these websites and actually uprank them so that they are better represented on a crawl. this is the project for the moment. It is a GitHub repository. so you would do a pull request and put URLs of the websites and then there are some classifications by topic even so you can contribute to this GitHub repository.

but I mean you can also contact us directly and share the URLs with us and we are working on making it a little bit more user friendly for people that are not familiar with git. we already run a pilot project with this and we work with masakanu which is African community of researchers mostly in natural language processing. they contributed 748 seats and we injected them over three consecutive crawls. We computed the web graphs and we managed to discover more than 28 million pages from those seats. So now to the second initiative which is improving language identification.

We run this project also collaborative project with the community. we already published a paper with 97 authors and is about building a language identification data set based on web data so that we can better evaluate the existing solutions out there. So okay why language identification? is because if you look at what's available, all of the models are very old and are ma high coverage models covering more than 200 languages are undervaluated and this is because of lack of data.

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So for the high coverage languages the only two available data sets for more than let's say 200 250 languages are either the declaration of human rights or the bible which is also there to a certain extent but that's not enough data we also have the question of what is the generalization across domains because again you only have three different sources and they are very specific. So we don't know how they generalize to other language registers. and then for our particular use case we are working with very noisy web texts and we need a robust classifier that can actually work with that. So in response to these problems we developed a crowdsourced data set based on common crawls data.

So we took two different recent s from common crawl. we also took samples from this data set that is also derived from common crawl called Madl 400. This is developed by Google. and then we recruited 80 native speakers. We asked them to validate the data that was annotated. We had to annotate the data to a certain extent because you cannot ask let's say a speaker of Swahili to reload documents until they find a Swahili document. so there was a little bit of bias in the annotation. This is the interface that they looked at. so they would encounter a web document already with the text extractor.

So they will only see the text and they will just select the text bands on their respective language and validate that the text is actually in the targeted language. They also had some optional annotations for future project. So asking if the text that they're seeing contains sexual toxic non-standard autotography or a boilerplate. for recruiting we work with two different communities work with Masakan that I already mentioned the African researcher community and with crowd which is the same concept but from people from Southeast Asia and then we built this data set called common lead so commonly supports 109 languages that's around 373,000

373,000 lines. you can see the language distribution there. So as you can see this actually targets a lot of those underrepresented languages. So it has good coverage in Africa and Southeast Asia. Sadly we found that it was extremely difficult to target languages in South America at least for the moment. But we are trying to get in touch with those communities as so you would think that people validating language in their own data would give you a very clean data set but that was actually not the case.

We had to do a lot of post-processing which included some automatic post-processing meaning cleaning the duplication filtering very short lines and also the web contains a lot of boiler plate so we have to check for contamination with English and then you do also a manual audit again for English contamination and then you have to reconcile multiple labels if you're working with multiple languages you would see that there are different standards for annotating those languages and they are not compatible with one another. So you have to reconcile those incompatibilities and you have also tried to reconcile things like there are macro languages and micro languages. So one example of that is Arabic and then all of the Arabic dialects and of course the web is very multilingual. So you will see code switching and you would have to do something with that. So make a decision.

in our case we kept it because I mean it's a feature of the internet. we evaluated some of these models that are more or less the standards that people use. so as you can see they have a very wide different range of coverage in languages different range of architectures and something that I want to highlight is that most of them don't actually tell you what data they were trained on. So it's very difficult to actually try to reproduce these models if you want to extend it to more languages. For instance, we evaluated in commonly set but also in other data sets. So you will see their bibles, the declaration of human rights,

Flores is based on Wikipedia. So these are some of the standard things that people already use for language identification.

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we well found out that this is actually a very hard task to do because again the models cover different sets of languages. So how do you even compare them fairly in your evaluation and the other one is that the different data sets that we're using here they use some of these different labeling systems that I told you are not compatible.

So again you have to standardize and reconcile them in order to have a comparison. And in our case we are trying to work with very large quantities of data. So we are also trying to take into account is the model accurate, but also is it fast is it going to work at the scale of common. So here are the results the F1 scores for a common lead.

So maximum score is 100 and decent score I would say for a task like this you would expect more than 90 and, as you can see for web data we are still very far. So you would see two different columns there. So this is because none of the models cover the 109 languages of common lit so the first one is evaluating over the 109 languages so if you don't cover the language you get zero for that language the second one is okay only for the language that the model covers we evaluate on that and we report the score and even on the ones that are covered by the models you can see in parenthesis how many are covered by each model. Even for that, you can see that the scores are not very good. So if you go from web data which is commonly to the other ones, you will find this problem and is that it doesn't even make any sense to try to report by data set because of the differences in So what we did is that we aggregated by sets of languages. So we have the high coverage So we only had two models that were covering more than a thousand languages. you can see the scores there.

Then you have the African languages that are extremely under represented in general.

Then we added Afrolith which is particularly targeted at these African languages and then you have the core languages that are the ones that are shared among all the models with the exception of Afrolit, because if you add that Afrolid model then the only language in common is Africans. So yeah that's the other problem. here for the core languages you can see that glot list does much better but then the question is gladly is a model that is trained on bibles and if you blend off all of the biggest evaluation data set is the one from Bible data.

So we did this Pareto curve, where you can see the Pareto frontier. So ideally you would like to be on the top very fast and very accurate. and the parto frontier is the line that you can see there. So if you are on that frontier that's the let's say best tradeoff for either speed or accuracy. So you can see that CLD2 which is a very old model developed by Google that's the one that we use internally at common crawl. it's actually very good because it's very fast.

So for our use case so far it's actually the best one compared to glot because gladly has more coverage and a little better accuracy but is much expensive for us to use.

And there's one question that people keep asking me when I say I work in this language identification task is like okay this sounds very easy why don't you use an LLM for this? there are two reasons. The reason is that if you compare something like LOIT to GPT models the open AI GPT models are significantly worse

that the very simple engram model. and the other reason is that for this evaluation we had to downsample and...

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create many versions of all of the data sets because otherwise it was extremely expensive to just do language identification with GPT models.

So they are not really a solution. these were the organizations that collaborated for this particular language identification effort.

And we have a leaderboard. If you are interested in and you think you can do better than any of these models here, you can submit a model to the leaderboard and we will evaluate it for you on our data set. And then we have code the package on Mozilla data collective. thank you. Yeah. How much better do you think? It depends for how many languages.

So I see so we have two issues and I think this is not isolated to language identification.

This is whatever task you can imagine in multilingual in general and the first problem is data. So it is very difficult to get data across more than 100 or 120 languages. you need a very very big effort from the community to try to do this. so you will always have that problem.

Especially if you want to do something let's say a thousand languages doing better than Bible data the declaration of human rights is almost impossible.

So that's the first issue.

The second issue is that we are hitting an architectural limit in the sense that when you have 100 200 250 task categories you can try to make a classifier.

The issue is that you can even use something like a hierarchical softmax fastex already use this and that will actually work a certain extent but when you go to 1,00 1,500 then I don't think we don't know how to do this properly yet.

That's one issue and the other issue is that for this specific task you also have the problem that there are many languages that are extremely similar to one another especially when you have dialectal data.

Distinguishing between them it's extremely complicated especially if you're mixing it with also other categories that are wildly different from one distinguishing when you have I don't know let's say Moroccan Arabic and Lebanese Arabic and then Chinese distinguishing from the macro language Arabic and Chinese is very easy but then if you made the general classifier that does all languages you try to then distinguish these two dialects it's very complicated. Yeah.

So dialinks are very complicated because a lot of them don't have standard autography.

We actually explored this to a certain extent. So this experiment is only working with textual data. but we had a little collaboration with Mozilla data collective. So they have their common voice data set. They have transcripts of in the common voice data set. and they even have let's say an informal speech data set and transcripts of that.

The issue is that one thing that we found out trying to train a model with them was that many of the dialects since they don't have standard orthography you would have transcriptions of exactly the same phrase in two different ways and that immediately complicates the task quite a lot. So I would say I mean I wouldn't dare to give you right now a specific results or specific numbers for that.

02:45:00

**Ronnie Falcon:** Okay. Hi. So, I'm Ronnie. I'm from Open Mined. Are we all sugar crashing? Are we doing well? I am. The main point that I wanted to bring in today is that the data that is going to be used to train the next frontier sovereign AI models is data that is currently not accessible or tapped by any frontier AI. And I think we made that point many times today. So hopefully by the end of the presentation we will reinforce that point.

It's great to see alignment across everyone in the community but also show how technically we can facilitate access to rare data through some of methods that developing at open mind is a nonprofit organization we're operating out of the US.

We are a community of people building software in the open source focused on unlocking access to private information in responsible value aligned way to society. we've been serving organizations like the ones on and we're a really passionate community. So please look at what we're doing if you're interested. Join us. And yeah, we're also hiring.

So if anyone is in the market so we made that base assumption in the past a couple of presentations ago but baseline assumption here is that for sovereign AI to be successful it needs to outperform private frontier models right it's not going to be better or adopted more if it's just local or open it needs to provide better capabilities than what it is out there

So a potential proposition is that access to rare data is a winning strategy right and this is not a very clever realization we've seen that many times ago in the past right that good AI needs good data and if we look at the timeline of kind of significant AI development moments that we've seen in the past it really corresponded to humanity's ability to gain access to quantities of larger new types of data. In 2012, ImageNet created the development of deep learning mechanisms. later, gaming environments allowed us to facilitate reinforcement learning models and more recently obviously the open web allowed us to develop large language models at the scale that we're seeing them today.

What AI labs claim, which is interesting, is that they're running out of data. and they're kind of already exploiting what is out there for them to learn from. But this is not a real claim. And I think our friend here mentioned that before. If we look at all the digitized content that exists out there, we're talking about 180 zettabytes of data. And if we look at the largest model that trained on the largest data set, we're talking about a 100 pabytes of data It's a good moment. Yeah. So, it's a drop in the ocean. It's really exciting to see. And obviously that 180 zettabytes of data represents a ton, millions of silos of data, not all connected, which represent all of our online activity, right?

So it's our financial records, online activity, preferences, business data, all of that, and obviously this is data that is not accessible for all the great reasons that we've discussed all today. privacy concerns, security issues, regulatory kind of rules, IP is a big one. so the next frontier of AI is data that is not on the public web, hopefully and it shouldn't be handed to a centralized provider like we've seen kind of slowly



emerging. The challenge can we build public interest frontier level AI while retaining data sovereignty? And hopefully we will kind of unpack how it could be done today even further.

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So, what if we could tap into a decentralized network of data holders? And that's kind of like the way that Openmind is thinking about solving for accessing all these data silos through a technology that we coined called In a network source AI paradigm, data owners have data sovereignty via a concept called attribution based control and data AI users can access these data and get better AI capabilities. In this model, data owners decide under which conditions they are willing to participate in a certain AI prediction.

They can be attributed basically we will say we will share this data if you name us. they can be paid or compensated or they can decide that they would like to enforce certain privacy technologies that will facilitate the access to the data in a way that is privacy preserving. So in reality the way that it looks is an AI user might tap into this network that we're describing. It will first discover the data on the network. It will then request usage over that data and that request is being sent to all the organizations to decide whether they're specifically interested to contribute to that specific data request. Some will say yes, but I'm charging XY Z for this data.

And once they approve, they would basically facilitate the request internally on premises and will return a result an AI driven result that could be then combined from all these sources into a prediction and an output that is shared with a user. We believe that these data organizations that hold private information will benefit from closed source AI assets.

they own the decisions under which their data is being used and they can man respond based on the data that they have to an AI query basically essentially meaning that they don't need to share the underlying data directly but just a insight of the data back to the open source asset that is being shared with a user. So this is what we mean by network source On one side we have data providers with On the other hand we have AI users looking for insights and we facilitate a network of individuals who work who could basically find the data and facilitate access to the data. So let's talk about a real world example that we have done recently and it's in the space of lo local journalism.

I will kind of just start by presenting that recently we are building the infrastructure to support 70 news organization in Indonesia who are all interested in allowing access to AI models through this interface and in a second you'll understand why it's probably much more beneficial for them in this kind of structure than what is happening today in the publisher space. So how it used to work, there were publishers that decided the terms under which they basically share access to their proprietary publishing data news articles and creative materials and on the other side there were readers who consumed the content that they prefer to use to read. AI kind of fundamentally broke that right.

So publishers no longer have the ability to get attributed for the data that they're providing. They lost compensation along the way and they lost visibility of what users are basically requiring access to their data because all of that is kind of being handled by the AI providers quering their data. On the flip side, we're also seeing sub-optimal experiences for the user, right? There is disinformation that can be emerged by the blackbox aggregation that we are seeing today. Value misalignment, right? Lack of trust and in some cases hallucination, especially when they don't have access to all of the data in specific, conditions or specific use cases. So why did publishers lose leverage and credit in AI systems? the first problem is copy problem.

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So once data is copied, it's gone, So handing over a copy of your content and you basically have no more say of how it's being used and then you're directly allowing others to compete with your services. The flip side, there is a problem of being data being blended in. You can't really trace it back. When an AI learns from your work, it gets mixed with millions of other sources and there is no technical way to point back and say this came from New York Times, So both copy and distillation problems are mechanisms by which we control transfers from publishers to platforms.

And it's very difficult to fix it under after the fact right tracing back and understanding which specific data piece of information was provided by specific provider obviously what I just shared is difficult to do and it's also really difficult to remove information after the fact and make a model unlearn something right so we believe that the answer is don't bake work into the model let the AI query content directly at inference time. So it can be directly attributed. This is very relevant in the context of the way data providers should negotiate their terms with current frontier models. Right? Tapestry adds a different flavor to this which I will touch on in a second.

So AI with attribution based control the mechanism that our software kind of enforces in the old way your work gets blended in and none can see where the answer came from and in this world the AI queries content live at inference time in the old system it's copied so you lose control and here you get structured access to insights not a copy of your files. so to tie it back to what we're doing in Indonesia, a user in the old model asks a AI model, hey, summarize the news. It goes into a black box, right? And spits out the results like here's a summary of all the news based on all the publications that I scraped in the past couple of hours.

And then in this model that we're basically suggesting the same user asking summarize the news there is an AI service that goes and requests information from all these news providers gets a synthesized response from them. They don't need to give them a copy of the news article, but they can just say, "Hey, here's the top things that happened today, and here is a summary, and then the AI kind of aggregator synthesizes it into a response for the user." so let's kind of tie it zoom even further out and tie it to how we think this type of technology could influence sovereign open frontier models and project tapestry specifically.

So this is kind of my attempt to describe what I understood from project tapestry where we have different sovereign nodes that all contribute to the training of shared base model right and then some sovereign derivatives that are doing fine-tuning for specific locations so that the response to the user is very localized and personalized right and there to that assumption is that the data that they can access to train these models is data that can be blended in and very people are okay with kind of that loss of control. and we see that this is working to some extent very well.

So this is a piece of research that I think was done by someone in this community showing that so they were basically sending out a survey val the world's value survey different location different organization different countries and they showed that different countries are plotted differently in its value progress in the two dimensions. So the Y axis people are either more traditional versus secular values and the survival versus values which basically corresponds to how wealthy a country is.

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And if you see so you can see how countries are kind of naturally plotted if you look in the black squares you'll see what values the frontier models are showing right so we can see and then I think the research so show showed that if you fine-tune a model you take the llama model that is represented here and you find it fine-tune it to Vietnamese culture, you can show that the distance actually from real Vietnamese value shrinks further. And I think as we do that, we do that better, this will shrink further.

But how can we bring bridge the gap the kind of like the last mile of improvement and that's where we believe that we can also derive insight from kind of data sets cultural archives that cannot be used directly in inference time in training time sorry and can only be accessible in inference. It could be because the data is just very timely in the case of publication and publisher data. It could also be because there is a reluctance to share the information without proper attribution. For example, I will share this data only if you pay me back give me a chunk of your value. so we believe that there is a very significant part of that 180t of data that would eventually only be accessed through this mechanism. This is essentially federated rag if people are familiar.

where the model itself queries at inference time different data sources and on top of that we basically apply a governance layer that determines under which conditions the data can leave the data sources in that inference moment. okay so if we're talking about a world where zetabytes of data is being siloed across different organizations institutions and a few frontier models kind of taking as much of it and developing a models for us. we actually envision that the future is an internet of private information right it's everyone's ability to connect to everyone and request structured access at

both on inference and training time but I'm just making a case more about the inference time so that we can learn from everyone's experiences in a way that is value aligned to kind of society and to the data owners so sovereign models drawing on sovereign data meaningful sovereign AI we get a lot the question of a technology is not a hard part here the hard part is how to incentivize people to do this I would say this is true. a lot of the work that we're doing at OpenMind is R&D on how to unlock deep incentives for data owners to share their data.

And we are developing technologies that help put the person who owns the data in the center and figure out ways that they're feeling comfortable or rewarded to share their data because we believe that this is the future of humanity. only if we unlock the access to private information, we will get to a point that we learn better and we learn from everyone's experiences, the second part that is challenging here is a social one and it's like we need a mutual orchestrator that will make everyone work together and kind of see the value of collaboration. So we're really excited to see project tapestry. We think that it's uniquely positioned to kind of work on that and yeah we are really excited to be part of it.

If you're interested in any of the things that are described here, please reach out or go into these further readings. and thank you so much. Nice to be here. On inference time. Yes. Yes. So, we have a proof of function with a few organizations like I said in the Indonesian media conglomerate that are all working with us on that.

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we are in active development on that. I think that there are some legitimate concerns regarding latency and scalability but those are the things that we're doing R&D and solving for. So the idea is to make it very

very production ready in the upcoming months. Yeah. Yeah. Yeah. So that's also an space for a lot of further R&D and then what we're doing. the good news is that aggregation and summarization is a very good LLM problems that we can do.

but it requires some enforcement on policy on how you present the results back to the user in a way that is attributed correctly and summarized and all of that. So yeah we have naive versions of it and doing R&D to improve on that. Thank you so much.

**Dean Wampler:** All right, we're in the home stretch here. Alexandra is going to talk about being successful in your AI journey. And there's still food and drink out there, too. We cannot leave until that's all gone.

**Alexandra Machado McDowell:** I'm standing between you and happy hour, right? I'm in between. So, I will try to make it entertaining.

So, we have heard about so many amazing topics. It has been a great afternoon. I am here to actually talk about something really different from what you have heard. It's actually how are we helping and hearing what are we hearing and seeing from customers around the world or how they're owning their AI journey and then also what does it mean for sovereignty. So it's very different from what you have heard u but very related.

So we're going to wrap up. That will be the last talk and it will be more applied. What are we really seeing in the field? So my name is Alexandra Mcdow Machalo. I'm part of the office of the CTO of Red Hat. I'm part I cover AI globally for a team called Fly Path. And before I actually kick it off, I want to share a little bit about our team so that you have a little bit of context of what I'm about to share. So we work from the office of the CTO with the customers and the CTOs and CIOs AC across top accounts from Red Hat. We hear their pain and we sit down with them and we say how is it looking for you in real life what are the issues that you're facing and with that we actually also support them.

So we just not hear only their griefs. we also support them to build a strategy and own right and hence the title own their platform AI journey and we do that with resources that I'm about to share with you and practical ways on how you can also bring it to your organization and also have that knowledge. So what are we hearing globally from our CTOs and our CIOs that we sit down and talk to? So a lot of them won't be surprising. So we have too many experiments and that's natural, Because we have been in experimentation mode for a couple of years and that's great.

But some organizations and companies have taken to the next level. And I'm here to share some examples of for example an oil and gas in the Middle East that has more than 7,000 projects in pilot base and they have a hard time getting to production because also costs and sovereign concerns. so both we have one of the biggest banks in France. they have 10,000 projects in pilot and they also have sovereign AI privacy concerns and they don't have a clear path to production and also safety right if we put that out there this agent that will talk to customers are we really sure that we can track everything we'll have explainability the answer is probably no then we also have the building new silos and this was maybe not a given right you say okay yeah we have

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a chief AI officer that just got dominated and we work with the top 10 banks across the world but in the US specifically they're having new teams being created data teams AI engineers which is fine they're

separate from the CTO and the CIO and they may reporting to the CEO and that's also fine but then they don't have a consistent strategy that goes across and then we have mentioned it also the tech stack right what does it look like also

also for AI and then last but not least and also that's why we're here and it brings everything home like the sovereignty concerns and this goes into geopolitical concerns right that we have right now but also data privacy and that's really big so for example one of the biggest banks in the world and they happen to be in Asia they have a lot of different location across the world and we are working with some of them and the main concern right there's hardware issues right they cannot buy hardware right now. So they need to work on the hybrid cloud, multicloud environments, but they are slowing down their AI adoption because of sovereignty concerns and data privacy. So we're seeing themes and these are the main themes that I wanted to bring up to you guys because it's appropriate and it brings me now right off the bat.

I'm going to share with you one of the resources that the fly path team has developed where we work with the CTO's and the CEOs and the chief AI officers and this we don't need to go into detail on each but I will give you an example on the end to end journey on their AI platform and why is this important because we have a lot of different vendors that work and frameworks and open source projects and different pieces of the puzzle.

So what we did this actually happened naturally by working with CIOs and CTO's across the world where we said we need to help them and what is missing they have all the products they have all I mean billions of dollars in investment what is the part of the puzzle that is missing so what we did this agnostic from any product we abstract from the product and the features and the frameworks and we work on capabilities end to end so what does it take to go

from exploring and being ready and I have data readiness. We need to talk with all the people that are there and we're always refining it as we test it with and as we're all evolving together to get to AI as a product. So just keep that in mind because I'm going to go right then into the solution. And what is the solution? You need to own your journey. You cannot let vendors first. The vendor lock is a bad idea especially also for cost and what is about to come.

even the hardware the rising prices it's a lot to handle so you need to be own it okay so that's the main thing now let's think about AI platform as a product has been there for so many years and I don't know if you guys have worked in tech I have worked for all my career and then AI platform as a product is not being talked about as much why because I meet with the chief AI officer and they're thinking about something else strategy how are the agents

is going to consume, they need to be personas. That's great, but we need to bring it down. It's tech. and this is where we're connecting the dots. So, this is what it give us is a clear path to production, right? You have all these pilots. You need to have a pave way to go from innovation to production and that's not a given. You need to work on those capabilities. The second is to build one foundation. So, it's okay to have different teams. That's natural. It happens. But you need to have one foundation where you can access the data sets all across many organizations. And you'll be surprised to hear that the biggest companies in the world don't have shared data sets. They don't have curated data sets that everyone could access and experiment with. and that's a problem, right?

So if we have that base of a shared data compute and more importance governance that would really give us that kind of end to end that we need to get to production and be effective but also have governance instill in there and that takes me to the last one that I have already repeated but I will do it again. It's

owning that end journey and that was give us a repeatable governor pattern across regions right for companies that are multinational but also across the hybrid cloud across you can go on multicloud and it will allow them to own it and this is super important I know we think about sovereignty a country also it involves a company owning their own destiny right and how they're doing their own technology and imple and actually doing their competitive advantage

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message right because at the end it will give them an edge. So I need to say this and I didn't include it in the other one because I wanted to highlight it and that's open source and that's what drives also our CTO right the office that I come from and also Red Hat. So open source is at the core of what we do. And open source it looks beautiful. Yes, it's there, but it's also more than that for the companies and organizations that we work with. When I talk to the CTO, CIOs, I make sure that they understand and we all know it because we're here that all the innovation it's happening at the speed of light because of open source. And if they don't own their journey and they don't have the flexibility for example, move models, right?

model as a service. You need to have options. a new model comes up in the other part of the world and it's better than anything. You need to have the flexibility to move quick to adopt new innovation and that is open source. So that's really important and I wanted to highlight it and it's actually a big part of the AI Alliance and why I'm here. So now I'm going to give you an example. Now I'm going to go into how are we doing this? This sounds great.

Okay, maybe I hope it does sound but how are we doing this? Okay, so this remember that puzzle that journey that I shared we convert So what we do is do it it becomes a maturity assessment based on capabilities that we run with those leaders and all the direct reports we actually work and this one is fresh from the oven. It's actually an innovation agency in a state in South America. So of a country and they're the innovation agency and we just run it a few weeks ago where we pick and we do it by column. So we say okay what are the columns that you need in order to get you at least started and what does this a self selfassessment.

So they actually selfassess where they're at in each capability and course we facilitate we're like are you sure you're in a one not a two or are you in a zero. So we have these conversations with them for this state agency governance by design that is over there it was very important right of because of the nature of what they do. So we pick this very important every customer's journey is different. Every organization too and every and it's a process. So this is the part also and this is very open source like you won't get there. I mean yes you can have a vendor and an out box solution now but it's not going to take you where you need because you're not following a process that you own.

So in this case I'm showing you examples of maturity assessments that we have done with different banks that are big part of who we support because they have a lot of money and then we are an investment and we go and they have a lot of innovation happening also at banks by and then here we have the recommended state. So what do we do with this AI journey? We tell them and let me go to the governance by design, right? And I will tell you how we run this examples of own your capability. governance by design, what does it mean for you, right? And this is how we do it. We run these workshops and we do it the oldfashioned way. And I'm going to tell you something that is really important that also relates a lot to open source. we do it with all the stakeholders that are touching that journey in the room.

So you will be surprised by knowing that a lot of these teams don't developers and I don't know if you remember the devops digital transformation where developers were not talking to the infra team. Now it's happening right with AI and data teams the data scientists don't talk to the platform team I'm not related to that but actually they provide the infrastructure platform that will allow them to innovate. So here we invite developers platform infrastructure very important CISO security the security teams are in silos too so we try to bring everyone in one room and that's where the magic happens and we tell them talk discuss where are you at but also it's a moment where they understand the capabilities that take to actually have AI delivered right so it also involves a lot of okay you need to know this there's

all of these underneath everything. So all of them are talking. So here I'm just going to explain just one of the boxes that is explainable and fair AI and as you could see they were in zero and we recommending you cannot go to one two you need to go to three right and these are the things that because we know we work with state agencies we work with banks we work with different domains we know hey you cannot afford to not have you need it and this is where you go and here are the recommendations from going

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from exploring zero to three. This need a step but also you need to start working on transitioning to advanced tools like chap and lime and you need to document your models with model cards and you need to do continuous evaluation eventually to get to that fairness metric on a recurrent schedule. So these are the recommendations that we give to them and we provide and something very important about this agency and you might be surprised my god their governance is so bad. you will be surprised to see that most of the customers that we're doing the AI journeys with, they're like this or worse. Okay, so actually this is pretty good how they're doing it. they also are going step by step. So they started with predictive AI and that's what they have in production because they could manage it and now they're starting to go into generative and agentic even though it seems like it's late. No, they're actually doing the right way.

they're taking it step by step and not doing 7,000 projects that can make it to production. So, we're gonna help them with that. And here are the recommendations in Spanish because they speak Spanish over there. And I just wanted to show you how in each capability we show them how that is done. So the key takeaways really from what we have learned in the field how AI is being applied and delivered by a lot of big companies that we work with. And I'm going to go back to how and the why.

And this is something that I already mentioned but I want to mention it again. AI couldn't be seen as a new thing like it's an evolution of technology and it needs to come with all the lessons learned that we have from our tech stack from all the years that we have gone into technology and we need to have that connection. So I always make sure that they like our CTOs and CEOs understand that connection and it's not a separate thing. We need to really work it together.

So what I already have said it but I will say it again it's owning your AI platform to the leaders building those capabilities and required across the end to end how and this is something really important that I have mentioned but that I maybe didn't go into detail we do value stream mapping where we have all the teams saying what piece and let's go to the data teams how do you ingest where are you working in

what systems are you using, is it in the company, who is accessing the data? So, we map the processes and the people that are touching end to end a specific value stream that we decide. and then that's really

important because it allows to align the people and the processes so that we can consolidate the tooling, right, and then the governance and create that share operating system needed to bridge those silos.

And then why it's gain that control and speed needed for that clear path right where production you get to production and you're not only getting to production you're doing it safely because that's the other thing right with governance and automation and things and explainability and all of the capabilities that you need to really control your sovereignty and something really important about why is that so innovation so clear path to production

working on these capabilities it's going to take so long I prefer doing outside of the box things not really if you actually do have a clear path and you work on those capabilities it will allow you to really innovate and also get to production because you will have the sandboxes and people can reproduce the projects so I deal with a lot of customers that have different teams doing chat boxes and they're not sharing information among the teams so they're kind of reinventing the wheel. So this is something important. Now I will end with a quote of one and it really brings it home about open source innovation and even sovereignty and what it takes and that's collaboration in my opinion even what we're doing here to me right of sharing information and knowledge is so important.

So this CTO from one of the biggest banks in Latam they said for the first time in the bank we brought together teams from technology, security, engineering, infrastructure, operations, data platform and development all in the same room aligned around a common objective. We found a space where collaboration was real, conversations were honest and we collectively began shaping a shared path forward. So I want to leave you with one thing from the field that I think we're doing here. as we're part of the AI Alliance and we're learning from each other and we're doing it through open source is that to really really get to the innovation levels but also the sovereignty and the levels of that we need to really do AI right we need to collaborate we cannot do it alone and bringing everyone in the room is really important and in a safe environment of course really important and that is what truly allows us to own our path right in AI

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and the AI journey. So yeah, so I want to that's it. Are you ready for happy hour? You know what? Yeah. Thank you.

mentioning that because actually part of the model architecture and strategy which we work as a capability with customers where we tell them you don't need to have all frontier models for everything predictive you can have small language models can have for different needs you need to be very targeted it's something that we also work so it was in my talking point but I get excited and I say other things so thank you for bringing it up thank you so yes it's something super important that we do work and it's an option and I don't think it gets enough attention and I think we'll get there.

**Dean Wampler:** Okay, we'll end the meeting now too for those of you online.

## Meeting ended

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